

QUESTION 1

What is the product of 573_8 and 246_8 ?

- A. 140958_8 B. 423236_8 C. 62914_8 D. 172702_8 E. None of these

QUESTION 2

Which of the following replaces **<*1>** in the code to the right to determine the number of command line arguments passed to `main()`?

- A. `String[].length`
 B. `String[].length()`
 C. `args.length`
 D. `args.lenth()`
 E. None of these

```
public static void main(String[] args) {
    int numargs = <*1>;
    // more processing not shown
}
```

QUESTION 3

Which of the following replaces **<*1>** in the code to the right to subtract the bitwise OR of `x` and `y` from the bitwise AND of `x` and `y`?

- A. `(x&&y) - (x||y)`
 B. `x - y`
 C. `(x&y) - (x|y)`
 D. `(x&&y) - (x^y)`
 E. None of these

```
public static int mystery(int x, int y) {
    return <*1>;
}
```

QUESTION 4

Assume that **<*1>** is filled in correctly. What is returned by `mystery(54, 90)`?

- A. 36 B. 108
 C. -36 D. -108
 E. None of these

QUESTION 5

Assume that the section of code marked with **<*1>** does not modify `i` or `j` and runs in time $O(n)$. What is the running time of the code to the right? Choose the smallest correct answer.

- A. $O(n)$ B. $O(n \log n)$
 C. $O(n^2)$ D. $O(n^3)$
 E. None of these

```
int n;
// code to initialize n not shown

int i=0, j;
while (++i<n) {
    j=i;
    while (++j<n) {
        <*1>
    }
}
```

QUESTION 6

Which of the following replaces each instance of **<*1>** in the code to the right to declare constant integer values accessible everywhere?

- A. `public int`
- B. `static final int`
- C. `public static final int`
- D. `final public int`
- E. More than one of these

For the remaining questions, assume **<*1>** has been filled in correctly.

QUESTION 7

Which of the following builds a `Map` named `m` which can be used to map from colleges to mascots?

- A. `Map m = new Map();`
- B. `Map m = new Map(College, Mascot);`
- C. `Map m = new HashMap(Mascot, College);`
- D. `Map m = new TreeMap(College->Mascot);`
- E. None of these

QUESTION 8

Assume that `Map m` has been built correctly, and that `College texas` and `Mascot longhorn` have been built correctly. Which of these adds to `Map m` the key `texas` with value `longhorn`?

- A. `m.put(texas.name, longhorn.name);`
- B. `m[texas] = longhorn;`
- C. `m.put(texas->longhorn);`
- D. `m.put(texas, longhorn);`
- E. None of these

QUESTION 9

Which of the following can be run outside `class Mascot` to check whether the mascot associated with `College osu` in `Map m` is an animal?

- A. `m.get(osu).isAnimal()`
- B. `m.get(osu).type == Mascot.ANIMAL`
- C. `((Mascot)m.get(osu)).isAnimal()`
- D. `((Mascot)m.get(osu)).type == ANIMAL`

```
public class College {
    // methods and constructors not shown

    private String name;
    private boolean publicSchool;
    private int enrollment;
}

public class Mascot {

    boolean isAnimal() {
        return type == ANIMAL;
    }

    // other methods and constructors not
    // shown

    private String name;
    private int type;

    <*1> ANIMAL = 0;
    <*1> HUMAN = 1;
    <*1> COLOR = 2;
    <*1> OTHER = 3;
}
```

E. More than one of these	
QUESTION 10 Which of the following replaces <*1> in the code to the right so that the method checks whether s1 is a subset of s2? A. s2.add(iter.next()) B. s2[iter.next()] == false C. s2[iter.next()] D. !s2.contains(iter.next()) E. None of these	<pre>public static boolean subset(Set s1, Set s2) { Iterator iter = s1.iterator(); while (iter.hasNext()) if (<*1>) return false; return true; }</pre>
QUESTION 11 Assume <*1> has been filled in correctly. Which of the following checks whether Set s1 and Set s2 have the same elements? A. subset(s1,s2) && subset(s2,s1) B. subset(s1,s2) subset(s2,s1) C. s1 == s2 D. s1.contains(s2).contains(s1) E. More than one of these	
QUESTION 12 What is returned by mangle("television")? A. television B. noisivelet C. tevnsisnse D. tensinfdel E. None of these	<pre>public static String mangle(String s) { StringBuffer sb = new StringBuffer(s); int len = sb.length(); for (int i=0; i<len; ++i) sb.setCharAt(i,sb.charAt((i*i)%len)); return sb.toString(); }</pre>
QUESTION 13 What is output by the code to the right? A. true -1 B. false -1 C. true 0 D. false 0 E. None of these	<pre>int x=0, y=0; boolean b = (++x < y) & (y-- > 5); System.out.print(b + " " + y);</pre>
QUESTION 14 Which of the following is not a subclass of Object? A String B. TreeSet C. Comparable D. Integer E. None of these	

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QUESTION 15

Which of the following replaces **<*1>** in the code to the right to make the `traverse()` method do a preorder traversal?

- A. `if (left != null) left.traverse();`
`System.out.print(data);`
`if (right != null) right.traverse();`
- B. `if (left != null) left.traverse();`
`if (right != null) right.traverse();`
`System.out.print(data);`
- C. `System.out.print(data);`
`if (left != null) left.traverse();`
`if (right != null) right.traverse();`
- D. `while (left) left.traverse();`
`while (right) right.traverse();`
`System.out.print(data);`
- E. None of these

For the remaining questions assume **<*1>** has been filled in correctly.

QUESTION 16

Which of these creates a BST and adds an object representing the integer 3?

- A. `BST b;`
`b.add(3);`
- B. `BST b;`
`b.new();`
`Integer i;`
`i.new(3);`
`b.add(i);`
- C. `BST b;`
`Integer i;`
`i=3;`
`b.add(i);`
- D. `BST b;`
`Integer i =`
`new Integer(3);`
`b.add(i);`
- E. None of these

QUESTION 17

Assume that n objects are added to a binary search tree in order from smallest to largest. What is the worst case running time for searching for an object in the tree? Choose the smallest correct answer.

- A. $O(1)$
- B. $O(\log n)$
- C. $O((\log n)^2)$
- D. $O(n)$
- E. None of these

```
public class BST {

    public BST(Comparable value) {
        data=value;
    }

    public BST add(Comparable value) {
        int c = data.compareTo(value);

        if (c<0) {
            if (right!=null)
                right.add(value);
            else right = new BST(value);
        }

        else if (c>0) {
            if (left!=null)
                left.add(value);
            else left = new BST(value);
        }

        return this;
    }

    public void traverse() {
        <*1>
    }
    // other methods not shown

    private BST left, right;
    private Comparable data;
}
```


QUESTION 18

Which of the following replaces **<*1>** in the code to the right to insert an element into the circular list after the current element?

- A.

```
CircleList c = new CircleList(o);
c.next = this.next;
c.previous = this;
this.next.previous = c;
this.next = c;
```
- B.

```
CircleList c = new CircleList(o);
c.next = this;
c.previous = this.previous;
this.previous.next = c;
this.previous = c;
```
- C.

```
CircleList c = new CircleList(o);
c.next = this;
c.previous = this;
this.next = c;
this.previous = c;
```
- D.

```
CircleList c = new CircleList(o);
c.next = this.next;
c.previous = this.previous;
this.next.previous = c;
this.previous.next = c;
```
- E. None of these

QUESTION 19

Assume **<*1>** has been filled in correctly. Which of the following replaces **<*2>** in the code to the right to print each element of the circular list exactly once?

- A.

```
while (cl != this) {
    System.out.print(cl.item);
    cl = cl.next; }
```
- B.

```
do {
    System.out.print(cl.item);
    cl = cl.next;
} while (cl!=this);
```
- C.

```
System.out.print(cl.item);
```
- D.

```
for(int i=0; i<cl; ++i)
    System.out.print(cl[i].item);
```
- E. None of these

```
// Creates a circular doubly linked list
```

```
public class CircleList {

    public CircleList() {}
    public CircleList(Object o) {
        item = o;
        next = previous = this;
    }

    public CircleList insert(Object o) {
        if (next == null) {
            item = o;
            next = previous = this;
        }
        else {
            <*1>
        }
        return this;
    }

    public CircleList remove() {
        // code not shown
    }

    public void print() {
        if (next != null) {
            CircleList cl = this;
            <*2>
        }
    }

    private Object item;
    private CircleList next;
    private CircleList previous;

}
```

QUESTION 20

Which of the following returns true?

- A. `mystery("banana")`
- B. `mystery("abcdeedcba")`
- C. `mystery("ASDFDSA")`
- D. `mystery("ananabanana")`
- E. More than one of these

```
public static boolean mystery(String s) {
    int len = s.length();
    for (int i=0; i<len/2; ++i) {
        if (s.charAt(i) != s.charAt(len-i-1))
            return false;
        if (s.charAt(i) > s.charAt(i+1))
            return false;
    }
    return true;
}
```

QUESTION 21

`GregorianCalendar` is a subclass of `Calendar` that is not abstract. Which of these statements is true?

- A. The `GregorianCalendar` class does not have to override the `add()` method and can not override the `set()` method.
- B. The `GregorianCalendar` class can not override the `add()` method or the `set()` method.
- C. The `GregorianCalendar` class must override the `add()` method but not the `set()` method.
- D. The `GregorianCalendar` class must override the `add()` method and the `set()` method.
- E. None of these

```
// The following is part of the built-in
// class Calendar

package java.util;

public abstract class Calendar implements
    Cloneable, Serializable {

    protected Calendar() {
        // code not shown
    }

    protected Calendar(TimeZone zone,
        Locale aLocale) {
        // code not shown
    }

    public static Calendar getInstance() {
        // code not shown
    }

    public abstract void add(int field,
        int amount);

    public final void set(int field,
        int value) {
        // code not shown
    }

    // other methods and class constants
    // not shown
}
```

QUESTION 22

Assume the `java.util` package has been imported. Where would the following declaration be allowed?

```
Calendar c = new Calendar();
```

- A. Only in package `java.util`
- B. Only in subclasses of `Calendar`
- C. Package `java.util` and subclasses of `Calendar`
- D. Anywhere
- E. None of these

QUESTION 23

Which of the following is possible as the return value of `s1.compareTo(s2)` if `s1` and `s2` are lower case words with `s1` coming before `s2` in the dictionary?

- A. -5
- B. 0
- C. 1
- D. 4
- E. None of these

QUESTION 24

Which of the following replaces **<*1>** in the code to the right to throw an appropriate exception?

- A. `throw Error();`
- B. `throw
 ArrayIndexOutOfBoundsException;`
- C. `throw RuntimeException();`
- D. `throw IllegalArgumentException;`
- E. None of these

QUESTION 25

Which of the following replaces **<*2>** in the code to the right to access the cost of the element in the list of components with index `i`?

- A. `((Component)comps.get(i).cost())`
- B. `((Component)comps.get(i)).cost()`
- C. `(Component)(comps).get(i).cost()`
- D. `comps.get(i).cost()`
- E. More than one of these

QUESTION 26

Suppose the class `Monitor` is an implementation of the interface `Component`, and that `Monitor` has a no-argument constructor. Which of these declarations is valid?

- A. `Component c = new Monitor();`
- B. `Monitor m = new Component();`
- C. `Component c = new Component;`
- D. `Monitor m = new Monitor;`
- E. More than one of these

QUESTION 27

What function is computed by `f()`?

- A. Addition
- B. Subtraction
- C. Multiplication
- D. Division
- E. None of these

```
public interface Component {
    public double cost();
    public double weight();
}

public class Computer {

    public Computer() {
        comps = new ArrayList();
    }

    public Computer addComponent(Component c)
    {
        if (c == null) <*1>
        else {
            comps.add(c);
            return this;
        }
    }

    public double cost() {
        double sum = 0.0;
        for (int i=0; i<comps.size(); ++i)
            sum += <*2>;
        return 2.0*sum;
    }

    private ArrayList comps;
}
```

```
public static int f(int x, int y) {
    if (x == 0) return 0;
    else if (x < 0) return -f(-x, y);
    else return y + f(x-1, y);
}
```

QUESTION 28

How many times is "sc" compared to an element in A when executing `find(A, 0, 5, "sc")` where A is the array below?

"aa"	"ac"	"db"	"ee"	"sc"	"zw"
------	------	------	------	------	------

- A. 0 B. 2
 C. 4 D. 6
 E. None of these

```
// Implements binary search
```

```
public static boolean find(Comparable[] A,
                           int r, int s,
                           Comparable item){
    if (r>s)
        return false;
    int mid = (r+s)/2;
    if (A[mid].equals(item)) return true;
    else if (A[mid].compareTo(item)<0)
        return find(A,mid+1,s,item);
    else
        return find(A,r,mid-1,item);
}
```

QUESTION 29

What is the maximum size of an array that can be searched with binary search using at most n comparisons with the item being searched for?

- A. $2n$ B. n^2
 C. 2^n D. $2^n - 1$
 E. None of these

QUESTION 30

Which of the following replaces `<*1>` in the code to the right to make it perform as specified?

- A. `sum = 0;`
 B. `return 0;`
 C. `System.exit();`
 D. No code is needed
 E. None of these

```
// Returns the sum of two integers
// represented as strings. If either
// string is not a number, returns the
// other number. If both are not numbers
// returns 0
```

```
public static int add(String s1, String s2)
{
    int sum = 0;
    try {
        sum += Integer.parseInt(s1);
    }
    catch(NumberFormatException e) {
        <*1>
    }
    try {
        sum += Integer.parseInt(s2);
    }
    catch(NumberFormatException e) {
        <*1>
    }
    return sum;
}
```

QUESTION 31

Assume `<*1>` has been filled in correctly. What is returned by `add("57", "-62")`?

- A. 0 B. -5
 C. 57 D. -62
 E. None of these

QUESTION 32

Which of these shows what array A looks like after the call process(A), where A is the array below?

17	-3	24	-5	10	10
----	----	----	----	----	----

A.

17	3	24	5	10	10
----	---	----	---	----	----

B.

0	0	0	0	0	0
---	---	---	---	---	---

C.

17	-2	24	-4	10	10
----	----	----	----	----	----

- D. A run time error occurs
E. None of these

```
public static void process(int[] A) {
    for (int i=0; i<A.length; ++i)
        if (A[i] < 0) A[i--]++;
}
```

QUESTION 33

What is output by output(7, 3)?

- A. 0 B. 1
C. 2 D. 3
E. None of these

```
public static void output(int x, int y) {
    if (x == 0) return;
    else {
        switch(y%4) {
            case 0: System.out.print(0);
                    output(x/10, y*y);
            case 1: System.out.print(1);
                    output(x/10, y*y);
            case 2: System.out.print(2);
                    output(x/10, y*y);
            case 3: System.out.print(3);
                    output(x/10, y*y);
        }
    }
}
```

QUESTION 34

What is output by output(27, 2)?

- A. 20 B. 012301230123
C. 2012330123 D. 20000
E. None of these

QUESTION 35

Which of these types would be a valid replacement for <*1> in the code to the right so that the assignment shown can be done without a cast?

- A. Integer
B. String
C. byte
D. long
E. More than one of these

```
int x = 27;
<*1> y;
y = x;
```

QUESTION 36

A valid line of input contains a student's first name, last name, and id number all separated by one or more spaces. What replaces `<*1>`, `<*2>`, and `<*3>` in the code to the right to extract this input?

- A. All are replaced with `in.nextToken()`
- B. All are replaced with `st.nextToken()`
- C. `<*1>`: `in.token(1)`
`<*2>`: `in.token(2)`
`<*3>`: `in.token(3)`
- D. `<*1>`: `st.getToken(1)`
`<*2>`: `st.getToken(2)`
`<*3>`: `st.getToken(3)`
- E. None of these

QUESTION 37

The `Student` class does not override the `equals()` method inherited from `Object`. When are two instances of `Student` equal according to the `equals()` method?

- A. When they have the same reference
- B. When their `firstname` fields have the same reference, their `lastname` fields have the same reference and their `id` fields are the same
- C. When their `firstname` fields are equal using the `equals()` overridden by `String`, their `lastname` fields are equal using the `equals()` overridden by `String`, and their `id` fields are the same
- D. Never
- E. Both A and B

QUESTION 38

What does matrix A look like after the call `rearrange(A)` where A is the matrix below?

1	2	3
4	5	6
7	8	9

- A.

1	4	7
2	5	8
3	6	9
- B.

7	8	9
4	5	6
1	2	3
- C.

1	2	3
4	5	6
7	8	9
- D.

3	2	1
6	5	4
9	8	7
- E. None of these

```
public class Student {

    // Assume IO.readLine() reads one line
    // of input from the keyboard and returns
    // it as a String

    public void input() {
        String in = IO.readLine();
        StringTokenizer st =
            new StringTokenizer(in);
        firstname = <*1>;
        lastname = <*2>;
        id = Integer.parseInt(<*3>);
    }

    // other methods not shown

    private String firstname, lastname;
    private int id;
}
```

```
public static void rearrange(int[][] A) {
    int[] temp;
    int[][] B = new int[A.length][];
    for (int i=0; i<A.length; ++i) {
        B[i] = A[A.length - i - 1];
    }
    A = B;
}
```

QUESTION 39

What replaces **<*1>** in the code to the right to select the substring of `big` starting at index `i` with length `smalllen`?

- A. `big.substring(i, smalllen)`
- B. `big.substring(i, i+smalllen)`
- C. `big.substring(i-1, smalllen)`
- D. `big.substring(i, i+smallen-1)`
- E. None of these

QUESTION 40

Assume **<*1>** has been filled in correctly. What is returned by `count("abracadabra", "abr")`?

- A. 2
- B. 3
- C. 4
- D. 5
- E. None of these

```
public static int count(String big,
                        String small) {
    int biglen = big.length();
    int smalllen = small.length();

    int total = 0;

    for (int i=0; i<biglen-smalllen; ++i)
        if (small.equals(<*1>)) ++total;

    return total;
}
```

Computer Science Answer Key

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1. D	11. A	21. C	31. B
2. C	12. C	22. E	32. E
3. C	13. B	23. A	33. D
4. D	14. C	24. E	34. C
5. D	15. C	25. B	35. D
6. C	16. E	26. A	36. B
7. E	17. D	27. C	37. A
8. D	18. A	28. B	38. C
9. C	19. B	29. B	39. B
10. D	20. B	30. D	40. A